

On the Structure of the Lattice of Noncrossing Partitions

Paper Summary - CHAU Dang Minh

The work focuses on properties enjoyed by the lattice of noncrossing partitions $\text{NC}(n)$, but not by the lattice of all partitions $\Pi(n)$, namely self-duality (which implies rank symmetry) and symmetric chain decomposition. Based on these results, several equalities are derived.

1 Introduction and Notations

A *noncrossing partition* π of $[n] = \{1, \dots, n\}$ is a partition of $[n]$ into blocks, such that whenever $1 \leq a < b < c < d \leq n$ and a and c are in the same block (later denoted as $a \sim_\pi c$) and b and d are in the same block, then a, b, c, d are in the same block. That means when drawing the numbers as points in a circle in clockwise or counter-clockwise order and connecting each pair in the same block with a chord, no two chords from distinct blocks intersect. Henceforth, $a \sim_\pi b$ if and only if there is a chord ab in the cyclic representation of π (or briefly, in π). For example, $\{\{1, 3, 8\}, \{2\}, \{4\}, \{5, 7\}, \{6\}\}$, from now on written briefly as $138|2|4|57|6$, is a noncrossing partition of $[8]$, while $138|24|57|6$ is not, as shown in Figures 1a and 1b, respectively.

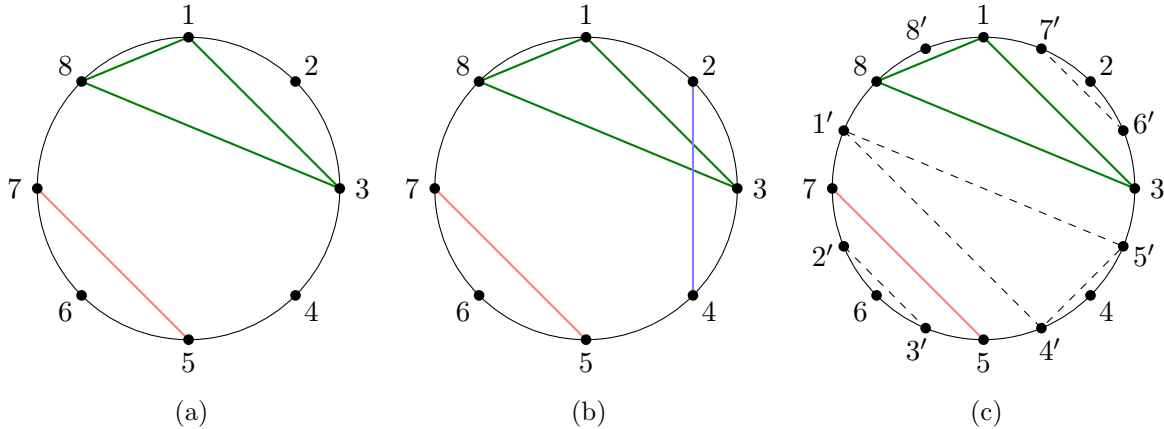


Figure 1

We also consider the linear representation, where the numbers are drawn on a line and a semicircle is drawn for each consecutive pair in the same block. For example, the linear representation of $138|2|4|57|6$ is shown in Figure 2. This allows us to speak of *nested blocks*. For example, in the same figure, the block 57 is nested in the block 138 .

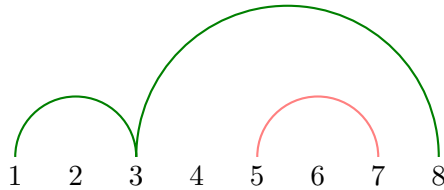


Figure 2

The number of noncrossing partitions of $[n]$ is the n -th Catalan number C_n . The number of noncrossing partitions of $[n]$ having i blocks is $W(n, i) = \frac{1}{n} \binom{n}{i} \binom{n}{i-1}$.

The *refinement order* on $\Pi(n)$, denoted by $\leq$, is defined by $\pi \leq \pi'$ if every block of π is contained

in a block of π' . For example, $138|2|4|57|6 \leq 1382|567$. We also denote by $\pi < \pi'$ the fact that π' covers π i.e. $\pi \leq \pi'$ and there is no partition π'' such that $\pi < \pi'' < \pi'$. That means π' is obtained from π by merging two blocks of π . The *rank* of a partition π is defined as $\text{rk}(\pi) = n - \text{bk}(\pi)$, where $\text{bk}(\pi)$ is the number of blocks of π .

2 Properties

Theorem 1. *For each $n \geq 1$, the lattice $\text{NC}(n)$ is self-dual. That is, there is an order-reversing bijection $\alpha : \text{NC}(n) \rightarrow \text{NC}(n)$ such that $\pi \leq \pi'$ iff $\alpha(\pi') \leq \alpha(\pi)$ for any $\pi, \pi' \in \text{NC}(n)$.*

Proof. Define α such that $i' < j'$ and $i' \sim_{\alpha(\pi)} j'$ iff there is no $i < j$ and $i \sim_{\pi} j$ with either $i' < n+1-j \leq j' < n+1-i$ or $n+1-j \leq i' < n+1-i \leq j'$. In the cyclic representation above, that means

1. Replace each i by $n+1-i$ for all $i \in [n]$.
2. Draw a point i' (it has value i , but distinguished from i) between i and $i+1$, with convention that $n+1$ is identified with 1.
3. Draw a chord between i' and j' iff it crosses no initial chord.
4. Revert the first step, i.e. replace each i by $n+1-i$ for all $i \in [n]$.

Let us show that $\sim_{\alpha(\pi)}$ is transitive. Let $i' < j'$, $i' \sim_{\alpha(\pi)} k'$ and $j' \sim_{\alpha(\pi)} k'$. Suppose that $i' \not\sim_{\alpha(\pi)} j'$. Consider the case that there is $i < j$ with $i' < n+1-j \leq j' < n+1-i$ (the other case is similar), or equivalently, $i' \leq n-j < j' \leq n-i$. Consider the cases

$$k' < i'; \quad i' < k' \leq n-j; \quad n-j < k' < j'; \quad j' < k' \leq n-i; \quad n-i < k',$$

we arrive at a contradiction in all cases to either $i' \sim_{\alpha(\pi)} j'$ or $j' \sim_{\alpha(\pi)} k'$. Therefore, the new chords correspond to the unique maximal non-crossing partition on $\{1', 2', \dots, n'\}$ whose chords do not cross the initial chords of π (an example of $\alpha(138|2|4|57|6)$ is shown in Figure 1c). Hence, α is well-defined. By definition of α , any chord of π does not cross any chord of $\alpha(\pi)$. Hence two statements $i < j \wedge i \sim_{\pi} j$ and $i < j \wedge i \sim_{\alpha^2(\pi)} j$ are both equivalent to

$$\exists i', j' : i' < j' \wedge i' \sim_{\alpha(\pi)} j' \wedge (i' \leq n-j < j' \leq n-i \vee i' \leq n-j < j' \leq n-i)$$

Therefore $\alpha^2 = \text{id}$ i.e. α is an involution, thus a bijection. Now let $\pi \leq \pi'$. Then $\alpha(\pi')$ does not cross π' , hence does not cross π , while $\alpha(\pi)$ is the maximal that does not cross π . Therefore, $\alpha(\pi') \leq \alpha(\pi)$. Thus, α is an order-reversing bijection, and $\text{NC}(n)$ is self-dual. $\square$

Theorem 2. *For each $n \geq 1$, the lattice $\text{NC}(n)$ is rank-symmetric. That is, the number of noncrossing partitions of rank k is equal to the number of noncrossing partitions of rank $n-1-k$ for any $0 \leq k \leq n-1$.*

Proof. Let α be as defined. Let $\pi < \pi'$, then $\alpha(\pi') < \alpha(\pi)$. Hence $\text{bk}(\pi') = \text{bk}(\pi) + 1$ and $\text{bk}(\alpha(\pi')) = \text{bk}(\alpha(\pi)) - 1$. Therefore, $\text{bk}(\pi') + \text{bk}(\alpha(\pi')) = \text{bk}(\pi) + \text{bk}(\alpha(\pi))$.

Now regard $\text{NC}(n)$ as a undirected graph. Since $12 \dots n \leq \pi$ for any π , the graph is connected and the previous equality holds for any π and π' . Since $\text{bk}(12 \dots n) = 1$, $\alpha(12 \dots n) = 1|2| \dots |n$ and $\text{bk}(1|2| \dots |n) = n$, we have $\text{bk}(\pi) + \text{bk}(\alpha(\pi)) = n+1$ for any π . Therefore, $\text{rk}(\pi) + \text{rk}(\alpha(\pi)) = n-1$. Hence, the number of noncrossing partitions of rank k is equal to the number of noncrossing partitions of rank $n-1-k$. $\square$

Recall that a symmetric chain decomposition (SCD) of a graded poset P is a partition of P into chains, such that each chain contains elements of ranks $k, k+1, \dots, n-k$ for some k .

Theorem 3. For each $n \geq 1$, the lattice $\text{NC}(n)$ admits a symmetric chain decomposition.

We outline the nonconstructive proof and focus on the constructive proof for later use.

Nonconstructive proof. Define R_i to be the set of noncrossing partitions such that the least element in the block containing 1 is i . We have $R_1 \cong \text{NC}(n-1)$ (seeing 1 as empty) and $R_2 \cong \text{NC}(n-1)$ (seeing 1 and 2 as a single point), hence $R_1 \cup R_2 \cong \text{NC}(n-1) \times \{0 < 1\}$. Also, $R_i \cong \text{NC}(i-2) \times \text{NC}(n-i+1)$ (identifying 1 with i , then split to two two independent circles $\{2, \dots, i-1\}$ and $\{i, \dots, n\}$). Using the fact that the product of two posets with SCDs also admits an SCD and checking the base cases, we conclude the result. $\square$

Constructive proof. We assign to each $\pi \in \text{NC}(n)$ a word $w(\pi) = w_1 \dots w_{n-1}$ over the alphabet $\{b, e, \ell, r\}$ as follows

$$w_i = \begin{cases} b, & \text{if } i \not\sim_\pi i+1 \text{ and } i \text{ is not the largest element in its block} \\ e, & \text{if } i \not\sim_\pi 1 \text{ and } i \text{ is not the smallest element in its block} \\ \ell, & \text{if } i \not\sim_\pi i+1, i \text{ is the largest element in its block} \\ & \text{and } i+1 \text{ is the smallest element in its block} \\ r, & \text{if } i \sim_\pi i+1 \end{cases}.$$

The letter b means there is a nested block of the one containing i that begins at $i+1$. The letter e means there is a nested block of the one containing i that ends at i . The letter ℓ means new non-nested blocks begins at $i+1$. The letter r means the block containing i continues at $i+1$. Let $B = \{i : w_i = b\}$ (respectively E, L, R).

Observation 4. We have $|B| + |E| + |L| + |R| = n-1$. Moreover, $1 + |B| + |L|$ counts the minimums in the blocks, while $1 + |B| + |E|$ counts the maximums. Hence $\text{bk}(\pi) = 1 + |B| + |L| = 1 + |B| + |E|$ and $|B| = |E|$.

Observation 5. Regarding each b in $w(\pi)$ as an opening parenthesis and each e as a closing parenthesis, we have a *complete parenthesization*. Indeed, if $w_i = b$, then there is a block containing i and has elements larger than i . Let j be the smallest such element, then $w_{j-1} = e$ and closes the parenthesis opened by w_i . Since each pair of parentheses corresponds to two successive elements in the same block, the parenthesization is complete.

Observation 6. Now we see that reading a letter b is going one deeper nested level, while reading a letter e is going one shallower nested level; reading a letter ℓ is starting a new block in the same nested level, while reading a letter r is continuing the current block. If we fix the positions of b and e , then every choice of ℓ and r in the remaining positions corresponds to a valid noncrossing partition. Therefore, the noncrossing partitions with the same B and E form a Boolean lattice isomorphic to $B_{n-1-2|B|}$, and these Boolean lattices are disjoint. Moreover, the correspondence $\pi \mapsto w(\pi)$ induces a bijection between the set of noncrossing partitions and a set of length- $(n-1)$ words in $\{b, e, \ell, r\}^*$ with the same number of b 's and e 's.

Now also regard the letters ℓ and r as opening and closing parentheses. Let ML and MR be the sets of positions of *matched* ℓ and r . We call $c = (B, E, ML, MR)$ the *core* of π . Once we fix that tuple, the minimal corresponding partition is when all the remaining letters are ℓ (deciding to exit a block as early as possible). Form a sequence of partition by replacing these $L - ML$ letters ℓ 's by r 's from left to right. We get a saturated chain. Using Observation 4, the number of blocks in the minimal partition is $n - |E| - |MR|$, while the number of blocks in the maximal partition is $1 + |B| + |ML|$. Since $|E| = |B|$ and $|MR| = |ML|$, the chain is symmetric. $\square$

3 Equalities

Corollary 7 (Touchard's identity). *For every $n \geq 1$, $\sum_{k \geq 0} \binom{n-1}{2k} C_k 2^{n-2k-1} = C_n$.*

Proof. We show that the left-hand side counts the number of noncrossing partitions of $[n]$. We count the number of noncrossing partitions with $|B| = |E| = k$. There are $\binom{n-1}{2k}$ ways to choose the positions of b 's and e 's, C_k ways to match them, and 2^{n-2k-1} ways to choose ℓ or r for the remaining positions. Summing over k gives the result. $\square$

Corollary 8. *For every $n \geq 1$ and $i \in [n]$, $\sum_{k \geq 0} \binom{n-1}{2k} \binom{n-2k-1}{n-k-i} C_k = W(n, i)$.*

Proof. The right-hand side is a known result of the number $W(n, i)$ of partitions having i -blocks. We count them in another way, again, for each k such that $|B| = |E| = k$. Using Observation 4, $i = 1 + |B| + |L| = n - |B| - |R|$ or $|R| = n - k - i$. There are $\binom{n-1}{2k} C_k$ ways to choose and match b 's and e 's and $\binom{n-2k-1}{n-k-i}$ ways to choose R . Summing over k gives the result. $\square$

Proposition 9. *Let $\overline{NC}(n)$ be the set of noncrossing partitions of $[n]$ with no consecutive elements in the same block and $NC_2(n)$ be the set of noncrossing partitions of $[n]$ whose blocks have size at most 2. Then $|\overline{NC}(n)| = |NC_2(n)|$.*

Proof. Consider the SDC in the constructive proof. Recall Observation 6 that the set of partitions with a fixed (B, E) is a boolean lattice. Since the b 's and e 's are matched, each boolean lattice corresponds bijectively to a partition in $NC_2(n-1)$, by seeing each pair of matched b and e as a block of size 2 and each remaining position in $[n-1]$ as a block of size 1. On the other hand, we count the boolean lattices by their minimum elements i.e. elements having $R = \emptyset$. Equivalently, $i \not\prec_{\pi} i+1$ for every $i \in [n-1]$. Hence, each boolean lattice corresponds bijectively to a partition in $\overline{NC}(n)$. Thus, $|\overline{NC}(n)| = |NC_2(n-1)|$.

A direct bijection can also be defined as follows. For each $\pi \in \overline{NC}(n)$, write the word representation of π , then let each matched pair of b and e correspond to a block of size 2, and each remaining letter correspond to a block of size 1. This gives a partition π' in $NC_2(n)$. Conversely, for each $\pi' \in NC_2(n)$, write the word w with $w_i = b$ and $w_j = e$ for each block of size 2 with minimum i and maximum j , and $w_k = \ell$ for each remaining position. This corresponds to a partition π in $\overline{NC}(n)$. $\square$

We also have a recurrence relation for $|\overline{NC}(n)|$, based on the smallest element $k \in [3, n]$ in the block containing 1, by identifying 1 and k and splitting the circle into two independent circles $[1, k-1]$ and $[k, n]$. We have, for each $n \geq 2$,

$$|\overline{NC}(n)| = \sum_{i=0}^{n-2} |\overline{NC}(i)| \cdot |\overline{NC}(n-i-1)|.$$

Note that $\overline{NC}(0) = \overline{NC}(1) = 1$. Let $f(x) = \sum_{n=0}^{\infty} |\overline{NC}(n)| x^n$ be the generating function of $|\overline{NC}(n)|$. Let $a_n = |\overline{NC}(n)|$, we have

$$\begin{aligned} f(x) &= a_0 + a_1 x + \sum_{n=2}^{\infty} \sum_{i=0}^{n-2} a_i a_{n-i-1} x^n = 1 + x + x \sum_{n=1}^{\infty} \left[\left(\sum_{i=0}^n a_i a_{n-i} \right) - a_n \right] x^n \\ &= 1 + x + x f(x)^2 - x f(x). \end{aligned}$$

Thus,

$$f(x) = \frac{1 + x - \sqrt{1 - 2x - 3x^2}}{2x}.$$